

Lab Exercise 3: Linear Regression

Evaluation: 5% of the total grade

This Lab problem involves the **Auto** dataset provided. In this dataset, **mpg** (miles per gallon) is the target feature or response variable. We want to develop a model for predicting **mpg** based on the descriptive features or predictors as input to the model.

Perform the following tasks:

- (a) For each predictor, fit a **simple linear regression** model to predict the response. Describe your results.
- (b) Fit a **multiple regression** model to predict the response using all of the predictors. Describe your results. For which predictors can we reject the null hypothesis $H_0 : \text{the effect of } d[j] = 0$?
- (c) How do your results from (a) compare to your results from (b)?
- (d) Create a plot displaying the univariate regression coefficients from (a) on the x-axis, and the multiple regression coefficients from (b) on the y-axis. That is, each predictor is displayed as a single point in the plot. Its coefficient in a simple linear regression model is shown on the x-axis, and its coefficient estimate in the multiple linear regression model is shown on the y-axis. Describe your observations on the plot.
- (e) Is there evidence of non-linear association between any of the predictors and the response? To answer this question, you may need plot each descriptive feature against the target or fit a model of the form below for each predictor $d[j]$:

$$M_w(\mathbf{d}) = \sum_{k=0}^b w[k] \times \phi_k(\mathbf{d})$$

Where $\mathbf{d}$ is a set of m descriptive features, $\mathbf{w}$ is set of b weights, ϕ_0 to ϕ_b are a set of basis functions that each transforms the input vector $\mathbf{d}$ in a different way.

Note1: Exclude potentially non-predicting descriptive features from the model building process. Briefly describe in the lab report which features were excluded and why.

Note2: You can use the entire dataset for training the model – skip the evaluation part for this lab exercise.

Note3: You can skip the detailed EDA for this lab – checks for data types and missing value should suffice.

Submit your lab report along with the source code preferably as Jupyter notebook in pdf format by June 17, 2025 @ 11:59.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5